

KEANU WATTS

PROFILE

Platform engineer and full-stack developer with a passion for infrastructure, game design, and building useful tools. Runs a personal Kubernetes cluster hosting 30+ services.

CONTACT

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WEB

theclusterflux.com

SKILLS

LANGUAGES

- Go
- TypeScript
- JavaScript
- Python
- C#
- Bash
- SQL
- Java

FRAMEWORKS

- React
- Next.js
- Express
- Gin
- Flask
- Angular
- Unity
- Tailwind

DATABASES

- MongoDB
- PostgreSQL
- MySQL
- SQLite

INFRASTRUCTURE

- Kubernetes
- Docker
- AWS
- Terraform
- Helm
- CI/CD
- GitHub Actions

MONITORING

- Grafana
- Prometheus
- Loki
- Tempo
- OpenTelemetry

TOOLS

- Keycloak
- Nginx
- cert-manager
- Kafka
- WebSockets
- MCP
- Git

LANGUAGES

English · Afrikaans

EMPLOYMENT

PLATFORM ENGINEER

Mar 2024 - Present

BBD Software Development, Cape Town

Managed AWS infrastructure for enterprise-scale client operations, balancing developer productivity with cost efficiency and system reliability. Led platform modernization initiatives including toolchain migrations and access governance. Maintained compliance with evolving industry standards and business requirements, and provided technical support to cross-functional teams.

SOFTWARE ENGINEER

Feb 2023 - Feb 2024

BBD Software Development, Cape Town

Working on event driven microservices on an AWS ecosystem, handling events from both the front end as well as other services via Kafka and websockets.

RPG DEVELOPER

May 2020 - Jan 2023

Momentum Metropolitan, Centurion

Developing and maintaining a newly built system for insurance policy management. Responsibilities included enhancement, defect investigation and new development for new business requirements.

PROJECTS

CLUSTER FLUX

Nov 2024 - Present

Personal Kubernetes cluster hosting diverse applications for personal and shared use. CI/CD, monitoring (Grafana / LGTM stack), and TLS via cert-manager.

LEARNHUB

Learning management platform with role-based access (student / instructor / admin), JWT auth, and MongoDB.
Go · React · MongoDB

ACCESS GUARD

Facial recognition-based access control with real-time monitoring and admin management interface.
React · Node.js · JWT

MCP SERVER – CLUSTER SERVICES

Model Context Protocol server enabling AI assistants to discover cluster services, inspect API schemas, and run read-only tests. Rate limiting and SSRF protection.
TypeScript · Kubernetes Client · Express

FINANCE SUITE

Debt and investment planning: loan comparison and visualization, investment target calculator, passive income planning with withdrawal-rate modelling.
Vanilla JS · Chart.js · Nginx

GAME DESIGN

Jun 2023 - Present

Tower Defense (Unity)

Pathfinding for dynamic maps, procedural difficulty scaling (Fibonacci), performance-aware tower placement and expanding levels.

Multiplayer FPS (Unity)

Custom gravity via quaternions, client-side prediction and lag compensation for smooth networked play.

Survivors-style (Go)

Entity-component architecture, 500+ enemies, spatial grids and object pooling for collision and frame stability.